

Robustness of Vision Algorithms Under Image Degradations

Digital Image Processing Course Project

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The Research Question

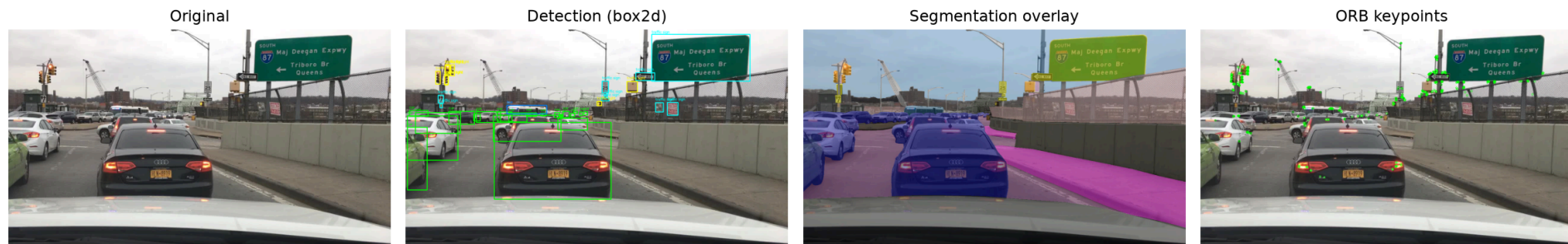
When dashcam images degrade, do **classical fixes** help, or do we need to **retrain the model**?

- We degrade clean driving images with three synthetic **distortions**: Gaussian noise, low light, and JPEG compression.
- We test two recovery strategies: **blind classical enhancement** (NLM / CLAHE / bilateral) vs **fine-tuning** the model on distorted data.
- We measure impact on three tasks spanning low-level to high-level vision: **ORB** feature matching, **YOLOv8n** detection, **SegFormer-b0** segmentation.
- Core tension: an image that *looks* cleaner is not necessarily a *better input* for a model.

Dataset: BDD100K

- Berkeley DeepDrive dashcam benchmark: diverse weather, lighting, and traffic.
- Robustness set: **100 images** (fixed seed 42) with **both** detection boxes and segmentation masks, so YOLO and SegFormer share an identical sample.
- Each image: RGB frame (1280×720), bounding boxes, and 19-class semantic masks.

BDD100K EDA: train split (1 sample)



Original · detection GT · segmentation overlay · ORB keypoints

Three Distortions, Four Levels Each

- **Gaussian noise:** target SNR (dB); levels 30 · 20 · 10 · 5 (mild to severe).
- **Low light:** gamma curve; levels $\gamma = 0.75 \cdot 0.5 \cdot 0.35 \cdot 0.2$ (progressively darker).
- **JPEG compression:** quality factor Q ; levels 90 · 50 · 20 · 10 (stronger blocking).

Intensity Sweep: Clean to Severe

Distortion intensity sweep - 440ae5e7-00000000.jpg

Gaussian Noise



Low Light



JPEG Compression



Distorted vs Enhanced

- Each distortion gets one matched classical enhancement: noise → **NLM**, low light → **CLAHE**, JPEG → **upscale + bilateral**.
- Shown at the strongest level: noise SNR 5 dB, low light $\gamma = 0.2$, JPEG $Q = 10$.
- Enhancement is **blind**: no access to the clean reference at inference.

Distorted → Enhanced (strongest level)

Distortion preview — 440ae5e7-00000000.jpg

Original



Clean image

Distorted



Target SNR: 5 dB
Measured SNR: 7.1 dB

Enhanced



NLM Denoising



Clean image



Gamma: 0.2
SNR vs clean: 4.9 dB



CLAHE



Clean image



JPEG quality: 10



Bilateral + Interpolation

Clean Baselines

Reference performance on the 100 undistorted images, before any degradation.

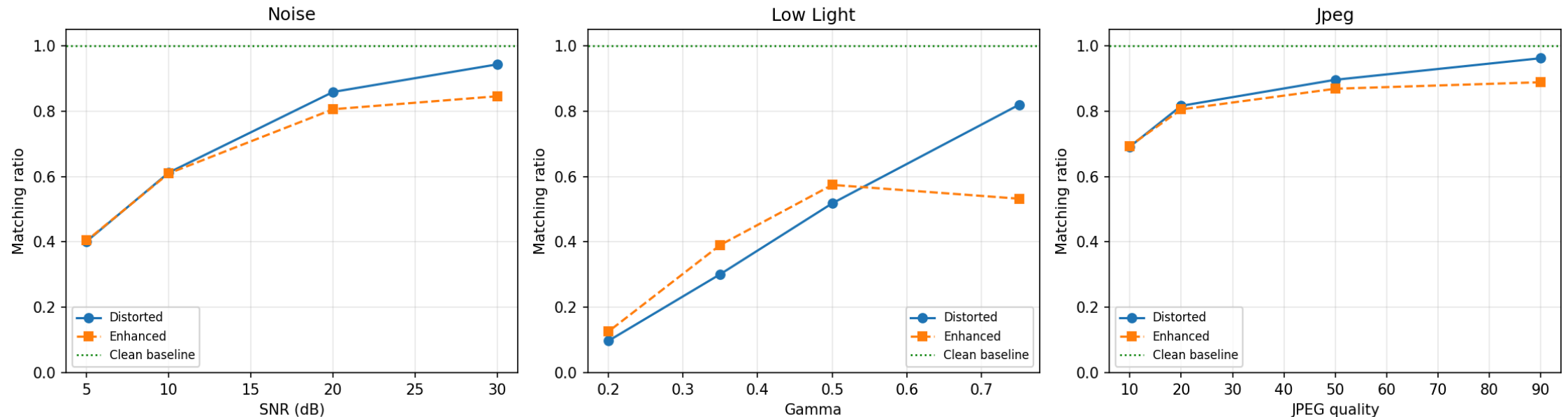
Task	Metric	Clean score
ORB matching	Match ratio	1.000
YOLOv8n	Recall @ IoU 0.5	0.319 (precision 0.725, matched IoU 0.813)
SegFormer-b0	Mean IoU	0.469

Trends on a 100-image subset, not official BDD100K leaderboard scores.

ORB: Fragile in Low Light

- Matching degrades smoothly with noise and JPEG, but **collapses in darkness**.
- Worst case $\gamma = 0.2$: matching ratio drops to **0.097** (from 1.000 clean).
- **CLAHE helps** low light: recovery **+0.089** at $\gamma = 0.35$, **+0.056** at $\gamma = 0.5$.
- **NLM hurts** mild noise (-0.098 at SNR 30 dB) by smoothing texture ORB relies on.

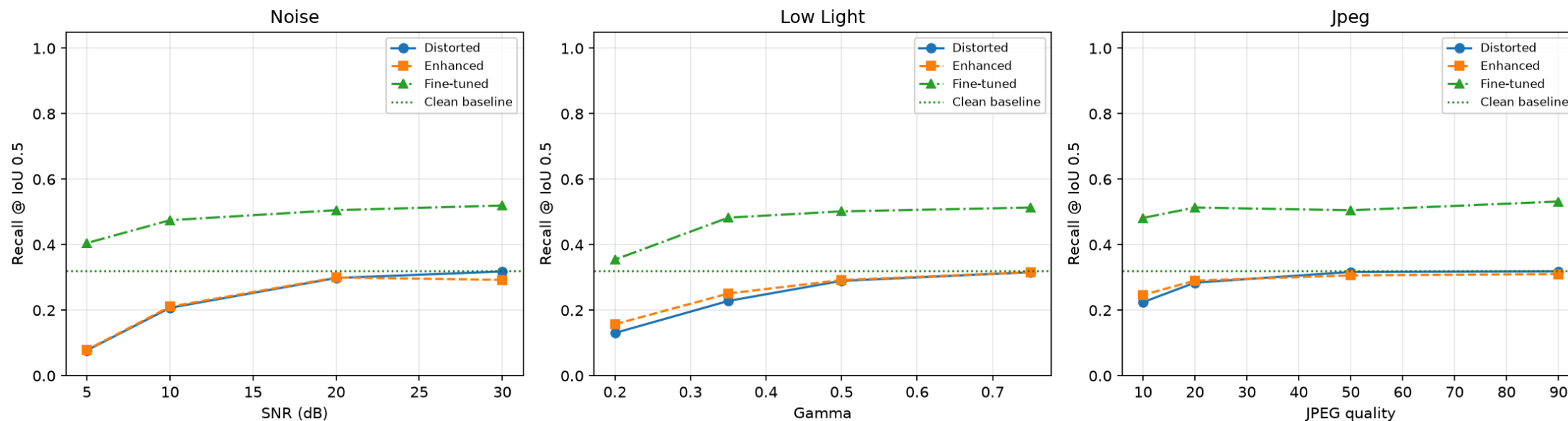
ORB matching ratio vs distortion intensity



YOLO: Collapses Under Noise

- Frozen YOLOv8n recall falls from **0.319** clean to **0.077** at SNR 5 dB.
- Mild JPEG ($Q = 10 \rightarrow 0.225$) and low light ($\gamma = 0.2 \rightarrow 0.131$) are tolerated far better than severe noise.
- **Enhancement barely helps:** NLM recovers only +0.002 at SNR 5 dB.

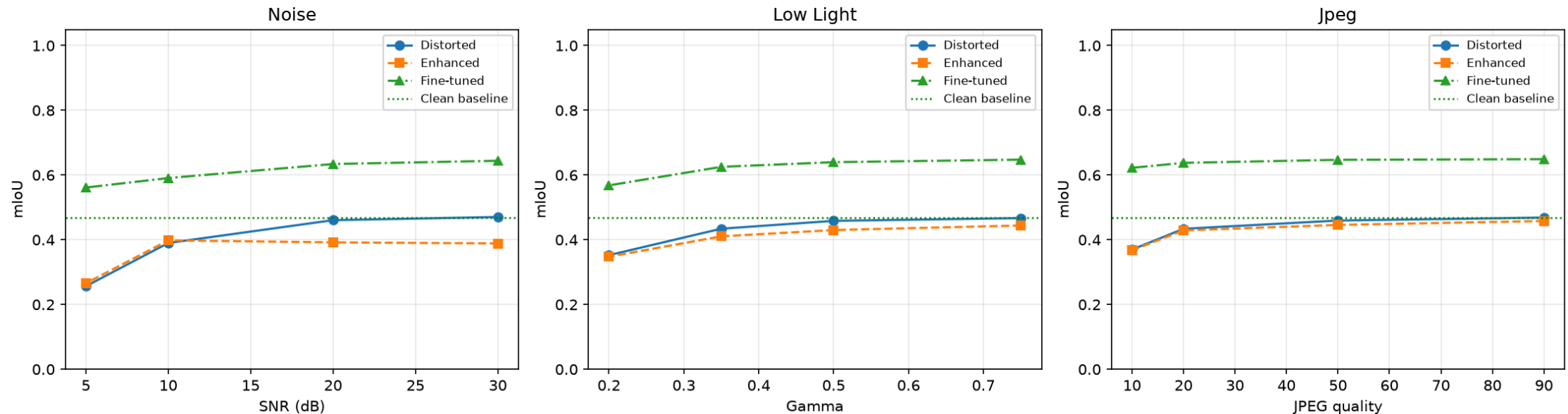
YOLOv8 detection recall vs distortion intensity



SegFormer: Noise Blurs Boundaries

- Frozen SegFormer-b0 mIoU falls from **0.469** clean to **0.257** at SNR 5 dB.
- Noise is the most damaging distortion: thin structures and class boundaries break down.
- **NLM even hurts** mild noise (SNR 30 dB: 0.470 $\rightarrow$ 0.389, gain -0.081).

SegFormer mIoU vs distortion intensity



Key Insight

A cleaner-looking image is **not** a better input for the model.

- **NLM** smooths texture and boundaries → hurts ORB descriptors and SegFormer.
- **CLAHE** lifts contrast for ORB matching, but does not help frozen deep models.
- **Bilateral** softens JPEG blocks visually but cannot restore discarded frequency content.
- Restoration quality and task-performance recovery must be measured **separately**.

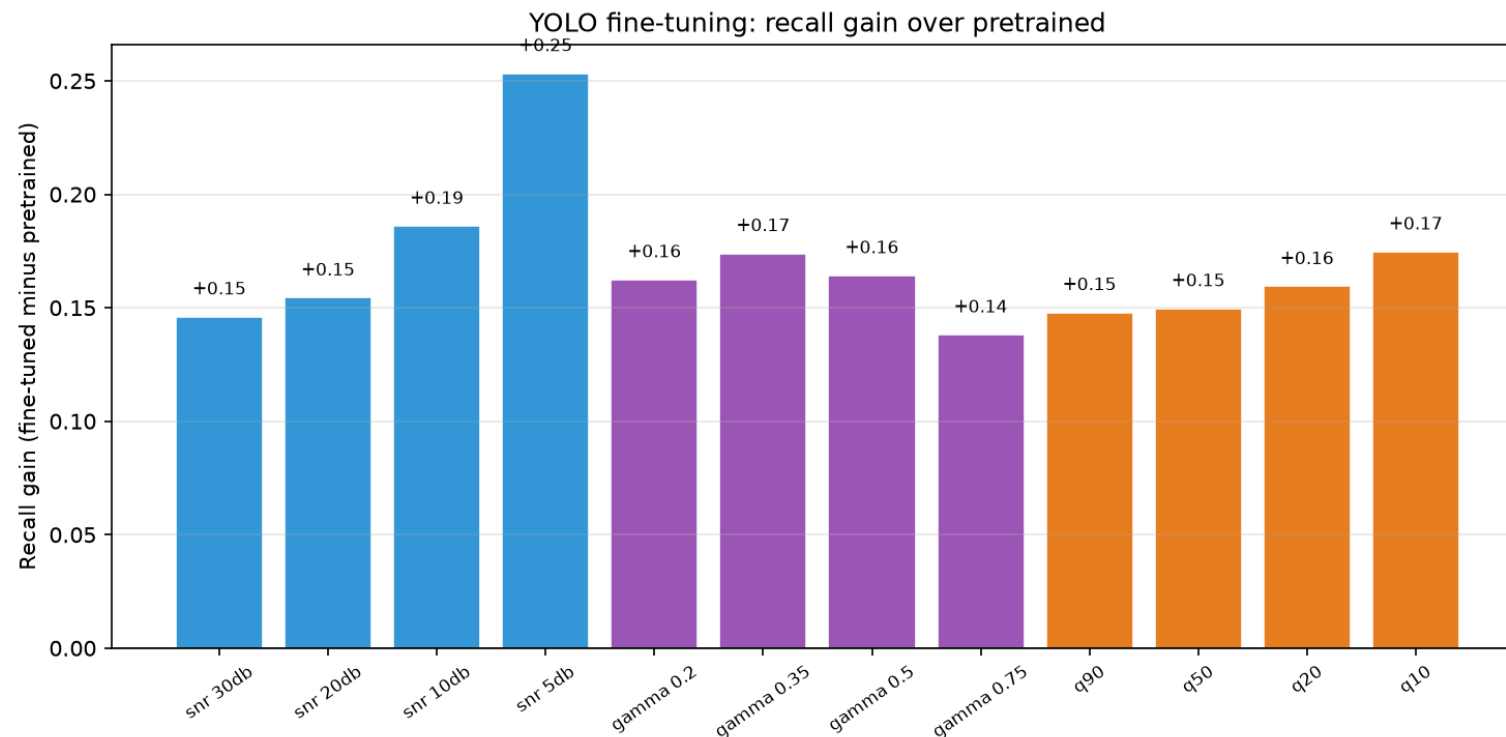
Fine-Tuning Fixes It

- Adapt the models by retraining on distorted data (500 train / 100 val per level, 30 epochs).
- Evaluated apples-to-apples on the **same 100-image robustness set** as the frozen models.

Task	Worst level	Frozen distorted	Fine-tuned
YOLO recall	Noise SNR 5 dB	0.077	0.405
SegFormer mIoU	Noise SNR 5 dB	0.257	0.561
SegFormer mIoU	JPEG Q = 10	0.348	0.622

Fine-Tuning: YOLO

- Fine-tuning improves recall at **every** distortion level (bars show fine-tuned minus pretrained).
- Largest gains hit the hardest cases: **+0.25** at SNR 5 dB, **+0.17** at JPEG $Q = 10$, **+0.17** at $\gamma = 0.35$.



Fine-Tuning in Action: YOLO

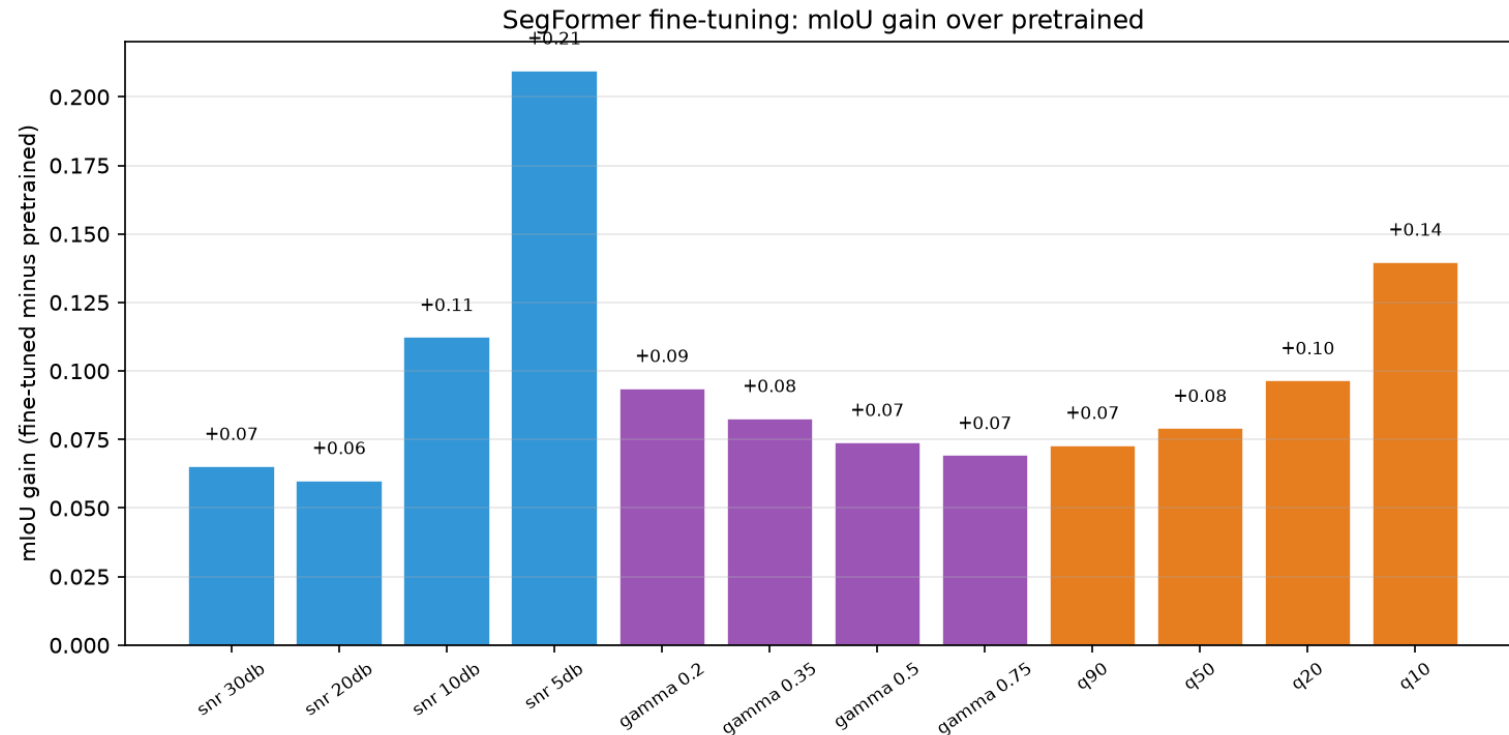
Noise SNR 10 dB: fine-tuned YOLO recovers the cars and van the pretrained model misses.



Left: pretrained · right: fine-tuned (green = detections).

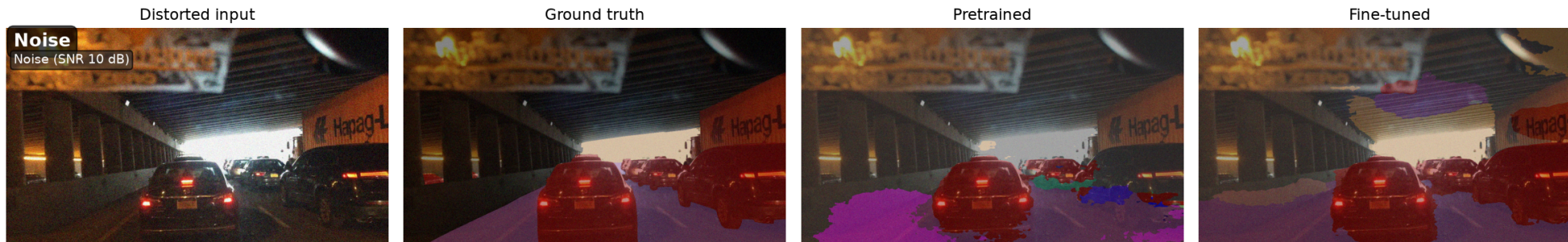
Fine-Tuning: SegFormer

- Fine-tuning improves mIoU at **every** level (bars show fine-tuned minus pretrained).
- Largest gains under severe degradation: **+0.21** at SNR 5 dB, **+0.14** at JPEG Q = 10, **+0.11** at SNR 10 dB.



Fine-Tuning in Action: SegFormer

Noise SNR 10 dB: fine-tuned SegFormer restores road, cars, and boundaries that the pretrained model loses.



Columns: distorted input · ground truth · pretrained · fine-tuned.

What Works When

Distortion	Best fix	Why
Gaussian noise	Fine-tuning	NLM smooths boundaries; adaptation learns the noise distribution
Low light	CLAHE (ORB) · fine-tuning (deep)	CLAHE lifts contrast for matching; deep models need adaptation
JPEG	Fine-tuning	Bilateral cannot restore lost frequencies; adaptation handles block artifacts

Conclusions

1. Frozen models fail under heavy noise; classical enhancement barely helps and sometimes hurts.
2. ORB gains from CLAHE in low light, but NLM/bilateral degrade its binary descriptors.
3. **Fine-tuning is the reliable fix** for YOLO and SegFormer across all three distortions.
4. Low-level and high-level robustness diverge: what looks cleaner is not what the model needs.

Thank You

github.com/NadavMoshe1/Image-Processing

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